GUQINAGENT: Tool-Interactive Performance-Action Planning for Jianpu-to-Jianzipu Generation

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Abstract

Jianzipu, the action-oriented notation of the Chinese guqin, jointly encodes strings, left-hand positions, right-hand strokes, ornaments, and stateful performance conventions. Generating it from a melodic score is therefore not mere text transduction: an output must agree with pitch, tuning, inherited state, and basic performability. We introduce GUQINAGENT, a tool-interactive formulation of constrained jianzipu generation. Our contributions are: (i) a full-score benchmark and executable runtime for iteratively editing a jianpu score into jianzipu; (ii) a warning-guided inference mechanism that feeds deterministic pitch and conservative playability diagnostics back to the agent without forcing a rewrite; (iii) a two-stage agent that separates pitch-grounded fingering from idiomatic guqin refinement; and (iv) targeted inverse trajectory construction that converts annotated scores into auditable, redacted tool-use supervision. Phrase partitioning is part of the generation algorithm, not an assumption on the input. The runtime exposes pitch candidates, stateful editing, compact context retrieval, and advisory warnings, and supports structured event-level evaluation rather than string overlap. We will release the model, redacted training trajectories, test benchmark, and executable runtime. This draft presents the task and system; final numerical results will be added after benchmark evaluation.

1 Introduction

Jianzipu is an open, compositional notation system that encodes how a guqin performance is produced, rather than merely which pitches sound. A notation unit can combine a string, hui position, left-hand finger, right-hand stroke, and ornament in one compact action description. Its interpretation is also stateful: harmonic mode remains active until terminated, walking gestures inherit positions, and compound gestures may cover multiple events. Similar text can thus imply different pitches, while different surface forms can be musically equivalent (Li et al., 2026b,a).

Recent work has made important progress on the preceding digitization problem: recognizing jianzi from manuscript images and recovering their internal compositional structure, including zero-shot component-sequence recognition and full-page mixed-script extraction (Li et al., 2026b,a). Our problem begins after such recognition. Given a machine-readable jianpu melody and score metadata, the system must select an executable realization in jianzipu. This is not transcription: the input does not prescribe a unique string, hui, finger, or ornament. It is a constrained performance-action planning problem, where pitch agreement is necessary but not sufficient.

Existing text generation alone does not expose whether a candidate realizes the intended pitch under the current tuning, inherits harmonic or left-hand state correctly, or remains compatible with notation coverage rules. Recent language-model research has explored interleaving reasoning with external actions, API use, and iterative feedback, but those methods do not model guqin-specific tuning, notation state, or performance constraints (Yao et al., 2023; Schick et al., 2023; Madaan et al., 2023). We formulate generation as interactive full-score plan editing. The model observes a complete jianpu score, partitions it into context-aware work units, queries deterministic tools when useful, and incrementally writes one event-indexed final notation plan. Pitch and state inconsistencies are exposed as actionable feedback, while idiomatic choices remain model decisions. In summary, our contributions are fourfold:

1. We introduce *constrained jianzipu generation*, a benchmark and executable runtime for reproducible score-to-jianzipu evaluation.

2. We introduce warning-guided inference: deterministic pitch and conservative notation-

constraint diagnostics become visible, actionable feedback rather than hidden post-hoc validation.

3. We design GUQINAGENT, a two-stage Fingering→Guqinizer agent that distinguishes basic sound production from stylistic refinement.

4. We develop targeted inverse trajectory construction, which replays annotations through the runtime and exports public, redacted supervision.

2 Related Work

Guqin notation and performance-oriented symbolic music. Recent work on jianzipu has focused on recovering structured notation from glyphs and manuscript pages (Li et al., 2026b,a). Earlier language-model-based work also studied transcription from jianzipu into modern notation (Shi et al., 2024). Our setting reverses this direction: starting from melodic events, the system must choose how to realize them on the instrument. A related problem arises in guitar tablature inference, where a pitch may admit several string–fret assignments and playability is assessed by musicians (Edwards et al., 2024). Work on piano fingering likewise models hand position and execution difficulty (Iman and Ahn, 2024). Jianzipu additionally includes state-dependent gestures and compact action notation that cannot be determined from pitch alone.

Tool-interactive planning and revision. ReAct interleaves reasoning and environment actions, while Toolformer and ToolLLM investigate tool-use learning and data construction (Yao et al., 2023; Schick et al., 2023; Qin et al., 2024). Self-Refine and Reflexion use iterative feedback to improve model behavior; CRITIC specifically studies revision driven by external tool outputs (Madaan et al., 2023; Shinn et al., 2023; Gou et al., 2024). GUQINAGENTuses a shared executable notation state to check pitch realizations and issue conservative, event-level warnings. These checks inform subsequent edits but do not determine musical suitability on their own.

Learning from interaction trajectories. Agent-Tuning and FireAct demonstrate the value of supervised interaction trajectories for language agents (Zeng et al., 2024; Chen et al., 2023). More closely related, RetroGen reconstructs process supervision from final expert artifacts for evidence-grounded generation (Huang et al., 2026). Our teacher likewise has access to a target artifact during offline construction, but is required to execute edits in the student-facing guqin runtime; the retained training context contains only public actions and observations. GEPA instead optimizes prompts by reflecting on execution feedback without updating model weights, providing a distinct comparison to trajectory fine-tuning (Agrawal et al., 2026).

3 Methodology

3.1 Constrained Generation with an Executable Runtime

Given a complete jianpu score $x = (m, e_{1:N})$, the task is to construct an event-indexed jianzipu plan $y = (y_1, \ldots, y_N)$. The metadata m specifies the mode and tuning; each event e_i has a stable index, notated duration, structural type, and a target pitch q_i when defined. An output entry y_i is either a jianzipu expression or an explicit empty display value. The latter may indicate continuation or coverage by a compound gesture and does not remove the event from the score. ABC notation is used only as an internal representation for pitch processing, not as an additional input.

Unlike independent token prediction, the interpretation of y_i depends on preceding actions. We represent this dependence by a deterministic notation-state transition,

$$s_i = \mathcal{T}(s_{i-1}, y_i; m), \qquad s_0 = s_{\text{init}}(m), \quad (1)$$

where s_i records such properties as harmonic-region status, inherited left-hand position, walking gestures, and compound-action coverage. Given this state, the realized pitch $\hat{q}_i = \mathcal{P}(y_i, s_{i-1}; m)$ is a partial function: some compound or ambiguous expressions do not admit a reliable pitch estimate. Structural consistency and pitch agreement are therefore distinct from the broader question of musical suitability.

Interactive plan editing. The complete score is partitioned into an ordered sequence of bounded, musically aligned work units $\mathcal{B}(x) = (p_1, \ldots, p_K)$. This partition is computed from the input score; the benchmark examples remain complete scores. The agent edits one shared plan rather than generating independent phrases. At interaction step t within p_k, its public history consists of the score, current

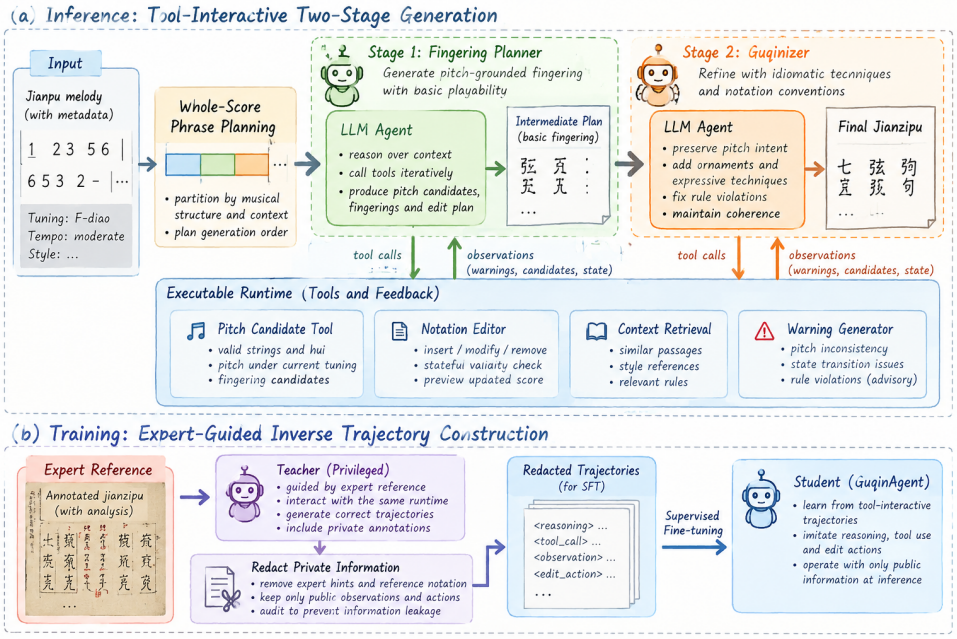


Figure 1: Overview of GUQINAGENT. **Top:** at inference time, the agent partitions a complete jianpu score into ordered work units, constructs a pitch-grounded intermediate plan in the Fingering stage, and refines it into final jianzipu in the Guqinizer stage. Both stages interact with the same executable runtime, which returns pitch candidates, plan previews, compact context, and advisory warnings. **Bottom:** for training, a privileged teacher uses expert annotations only through a private channel while interacting with the public runtime. Redaction retains a student-visible trajectory for supervised fine-tuning, so the deployed student conditions only on score context, its prior actions, and runtime observations.

work unit, editable plan, replayed notation state, and previous tool observations:

$$h_t = (x, p_k, y^{(t)}, s^{(t)}, o_{<t}),$$
$$a_t \sim \pi_\theta(\cdot \mid h_t, r), \tag{2}$$

where r denotes the current planning stage and $s^{(t)}$ is obtained by replaying the current plan. The executable runtime $\mathcal{R}$ processes an action and returns the resulting plan, state, and tool observation:

$$(y^{(t+1)}, s^{(t+1)}, o_t) = \mathcal{R}(x, y^{(t)}, s^{(t)}, a_t). \tag{3}$$

Read-only actions leave the plan unchanged. The runtime used to construct training trajectories is also used at inference time.

The agent can request tuning-aware string–hui realizations through `get_pitch_candidates`, including open strings, stopped notes, harmonics, and supported inherited or compound gestures. Returned candidates distinguish exact, approximate, and unavailable matches. The only plan-writing operation is `edit_plan(jianzi_rows)`, which applies event-indexed updates and replays the resulting notation state. Compact context retrieval through `list_context` and targeted earlier-phrase lookup provides prior material without reproducing the entire score in every interaction.

Conservative constraint feedback. We distinguish runtime-checkable violations from cases whose interpretation is uncertain. For a rule c and event i, let $v_c(i; x, y, s) \in \{0, 1, \perp\}$ denote no detected violation, a detected violation, or an indeterminate check, respectively. The warning set is

$$W(x, y) = \{(i, c) : v_c(i; x, y, s) = 1\}, \tag{4}$$

with s obtained by replaying y. In the implementation, checks cover reliably parsed pitch mismatches and conservative diagnostics for harmonic-state continuity, walk direction or span, interrupted open-string resonance, paired stroke adjacency, and difficult left-hand reaches. Indeterminate cases are not converted into violations. Warnings accompany the relevant `edit_plan` preview but neither rewrite the plan nor imply that an unflagged output is musically correct. The agent may query candidates, revise the affected entries, or retain its choice; identical repeated edits are stopped to prevent unproductive loops.

3.2 Two-Stage Performance-Action Planning

The same notation event jointly specifies how a pitch is produced and how it is articulated. We separate these decisions into two successive planning stages. *Fingering* first constructs a basic pitch-grounded realization, choosing strings, hui positions, left-hand contact, and elementary right-hand strokes. *Guqinizer* then revises the committed plan for idiomatic execution, including ornaments, walking gestures, harmonic transitions, and compound-action notation. The second stage may preserve or revise earlier decisions; it is not a post-processing formatter.

For work unit p_k, let $y^{(k-1)}$ be the shared plan after completing the preceding units. A bounded runtime rollout under each stage gives

$$\begin{aligned} y^{F,(k)} &= \mathrm{Rollout}_F(x, p_k, y^{(k-1)}; \mathcal{R}), \\ y^{(k)} &= \mathrm{Rollout}_G(x, p_k, y^{F,(k)}; \mathcal{R}), \end{aligned} \tag{5}$$

where each rollout consists of stage-conditioned tool interactions of the form in Eq. (3). Only edits committed through the runtime enter the shared plan. Crucially, the state passed to p_{k+1} is replayed from $y^{(k)}$, including any earlier mistakes, rather than substituted from an expert reference. Both stages retain access to pitch candidates and runtime feedback. Warnings can trigger another revision but are not hard acceptance criteria; execution remains subject to the configured interaction budget.

This decomposition imposes a procedural bias: establish a basic sound-producing realization before making the more context-dependent choices involved in idiomatic execution. It does not assume a unique separation between fingering and ornamentation, nor does it guarantee that the second stage preserves every correct decision. We evaluate both the benefits and regressions introduced by refinement against a single-stage agent using the same public tools.

3.3 Expert-Guided Inverse Trajectory Learning

Expert jianzipu annotations specify desired performance actions but generally not the tool interactions by which an agent could arrive at them. Directly supervising final notation therefore leaves the interaction policy unspecified. We instead construct trajectories by allowing an offline teacher to work backward from the annotated target while acting through the same executable runtime available to the student.

Let z be a phrase-level expert annotation and $\mathcal{K}(z)$ the technique-relevant descriptions retrieved from a curated knowledge base. Both are private to the teacher. Its rollout is conditioned on the input, private guidance, and public runtime:

$$\tau^T \sim \mathrm{Rollout}\big(\pi_T \mid x, z, \mathcal{K}(z); \mathcal{R}\big), \tag{6}$$

where τ^T records the chronological sequence of tool actions, returned observations, and accepted plan updates. The reference guides the search for an executable realization but does not constitute a tool response. The teacher constructs stage-specific traces: basic fingering first, followed by idiomatic refinement. Semantic descriptions clarify technique constraints that may be implicit in compact jianzipu symbols; they are not added to the student's inference context.

Validation and public-prefix reconstruction. Candidate examples are screened for reference parsing and consistency before construction, and generated traces are checked against the runtime. Training examples are obtained by removing private expert content and retaining the public interaction sequence,

$$\begin{aligned} \tilde{\tau} &= g_{\mathrm{pub}}(\tau^T), \\ \mathcal{D} &= \{\tilde{\tau} : \tau^T \text{ passes validation}\}. \end{aligned} \tag{7}$$

The intended information boundary is temporal as well as structural: at each supervised action, the conditioning context should contain only the input, earlier public actions, and observations already returned by the runtime. Removing private fields alone cannot establish this property if rewritten reasoning anticipates future observations or reproduces unseen annotation details. Public-prefix consistency must therefore be checked separately from runtime execution validity; the precise checks and filtering counts belong in the experimental protocol.

Student supervision. We fit a stage-conditioned student policy to the assistant-side actions and messages in the retained public trajectories. For a supervised token position u, let h_u^{pub} contain the preceding student-visible context and let w_u be its target token. The training loss is standard masked cross-entropy,

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\sum_{\tilde{\tau} \in \mathcal{D}} \sum_{u \in \mathcal{Y}(\tilde{\tau})} \log p_\theta(w_u \mid h_u^{\mathrm{pub}}), \tag{8}$$

Component	Contents
Input	Complete-score metadata, tuning, jianpu events, and stable event indices.
Output	One full-score, event-indexed jianzipu plan with explicit empty display values.
Runtime	Pitch candidates, plan editing, compact context lookup, validation, and audit.
Reference	Expert annotation, private to the offline teacher and evaluation pipeline.

Table 1: Benchmark representation. The student policy never receives the reference plan.

Method	Pitch Acc. ↑	MAE ↓	String ↑	LH F1 ↑	RH F1 ↑	Rules ↓
Direct	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
Single-stage agent	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
Two-stage GUQINAGENT	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]

Table 2: Results template for the high-confidence test subset. Final numbers will include denominators and unresolved-event counts.

Inference feedback	Pitch Acc. ↑	Rule Viol. ↓	Warning repair ↑	Repeated-edit stop ↓	Mean turns ↓	Expert playability ↑
Without warning feedback	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]
With warning feedback (GUQINAGENT)	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]	[TODO: –]

Table 3: Warning-guided inference ablation. Both rows use the same model, tools, decoding configuration, and maximum turn budget; only runtime warning feedback differs. "Warning repair" is the fraction of initially warned events that are later changed and no longer warned. "Repeated-edit stop" counts trajectories terminated after an identical edit submission. Expert playability is reported only if a blinded musician study is conducted.

where $\mathcal{Y}(\tilde{\tau})$ indexes supervised assistant tokens; tool outputs form context but are not prediction targets. At deployment, neither expert annotations nor private semantic guidance are provided. The resulting student must recover its actions from the input score, its own previous edits, and the executable feedback it has observed.

4 Experiments

4.1 Experimental Setup

Benchmark and Data Protocol. The benchmark uses score-level splits. A high-confidence test subset retains scores with reference pitch-match rate strictly above 70% under the current 50-cent auditor, calculated over parseable reference events only. The current test inventory has 44 scores; 39 meet this condition. The other scores remain available for diagnostic analysis rather than being silently treated as clean gold pitch labels.

Evaluation Metrics. We parse predictions and references into unified events containing source and event IDs, target pitch, string, left-hand position and finger, right-hand technique, tone type, ornaments, and necessary state. We do not use BLEU, ROUGE, or direct final-string equality as primary metrics. Let $\hat{q}_i$ and q_i^{ref} denote the pitches realized by the prediction and the reference at event i, let $\delta_i = d_c(\hat{q}_i, q_i^{\text{ref}})$ measure their distance in cents, and let $\mathcal{I}_{\text{pitch}} = \{i : \hat{q}_i \text{ and } q_i^{\text{ref}} \text{ are both resolvable}\}$ be the jointly resolvable set. Our primary pitch metrics are

$$\begin{aligned} \text{PitchAcc}_{50} &= \frac{1}{|\mathcal{I}_{\text{pitch}}|} \sum_{i \in \mathcal{I}_{\text{pitch}}} \mathbf{1}[\delta_i \le 50], \\ \text{PitchMAE} &= \frac{1}{|\mathcal{I}_{\text{pitch}}|} \sum_{i \in \mathcal{I}_{\text{pitch}}} \delta_i. \end{aligned} \tag{9}$$

Unresolved events are excluded from the denominator rather than counted as errors, and $|\mathcal{I}_{\text{pitch}}|$ is reported alongside every pitch number. Beyond pitch, we report string accuracy; left- and right-hand fingering precision, recall, and F1; ornament precision, recall, and F1; deterministic rule-violation rate; and technique-usage statistics to expose collapsed or overused techniques. Each violation is stored with an event ID, relevant state, and explanation. Rules that cannot yet be determined reliably are excluded from automatic scoring and listed as expert-review TODOs.

4.2 Results and Discussion

Comparisons and Ablations. We will compare direct generation, a single-stage interactive agent, and the two-stage GUQINAGENT. Ablations will remove pitch candidates, warning feedback, stateful audit, or compact context retrieval. We ask whether tools improve pitch correctness and rule compliance, whether warning feedback improves repair behavior without inducing unproductive loops, whether stage decomposition improves idiomatic notation, and whether inverse trajectories improve performance without simply increasing tool length.

5 Conclusion

GUQINAGENTframes jianzipu generation as executable, stateful plan editing. Its benchmark makes score events, tuning, and notation state explicit; its two-stage architecture separates basic realization from idiomatic refinement; and its inverse trajectories create auditable public tool-use supervision from private annotations. Together, these components enable evaluation beyond text overlap and establish a practical basis for musically grounded

notation agents.

6 Limitations, Ethics, and Resource Availability

The auditor is conservative and incomplete: no warning is not proof of musical correctness, and unresolved notation is not a success. Fine-grained left-hand reachability, timbre, and lineage-specific style still require expert review or additional physical modeling. The digitized score collection may not represent all historical sources or playing traditions. We position GUQINAGENTas an assistive research system, not a replacement for performers or tradition bearers. Before release, we will review source licenses and provenance, remove private teacher material, and publish only redacted trajectories intended for public supervision.

Open Science and Resource Availability. We will release the trained model, redacted training trajectories, the test benchmark, and the executable runtime used by teacher generation and inference. The release will include score-level split definitions, event-indexed schemas, evaluation scripts, filtering manifests, and instructions for reproducing the benchmark. Private teacher prompts, unredacted teacher reasoning, and any source material that cannot be redistributed will not be included. The archival repository URL, model license, dataset license, and versioned release identifier will be added before submission: [TODO: repository URL and licenses].

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